



AFRL

DBOL – INTERACTIVE TOOL FOR ONTOLOGY BUILDING

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BLUF

- Creating ontologies is a time and labor-intensive process. How can we shift some of the burden to the SMEs?
 - Without compromising on the quality of the ontologies
 - Ensuring that the ontologies are scoped correctly
- Our approach is an ontology development tool DBOL that guides the SME by asking questions and relying on axiomatization to create valid ontologies



Use Cases



Create new domain ontology using SME knowledge



Create new domain ontology using authoritative sources

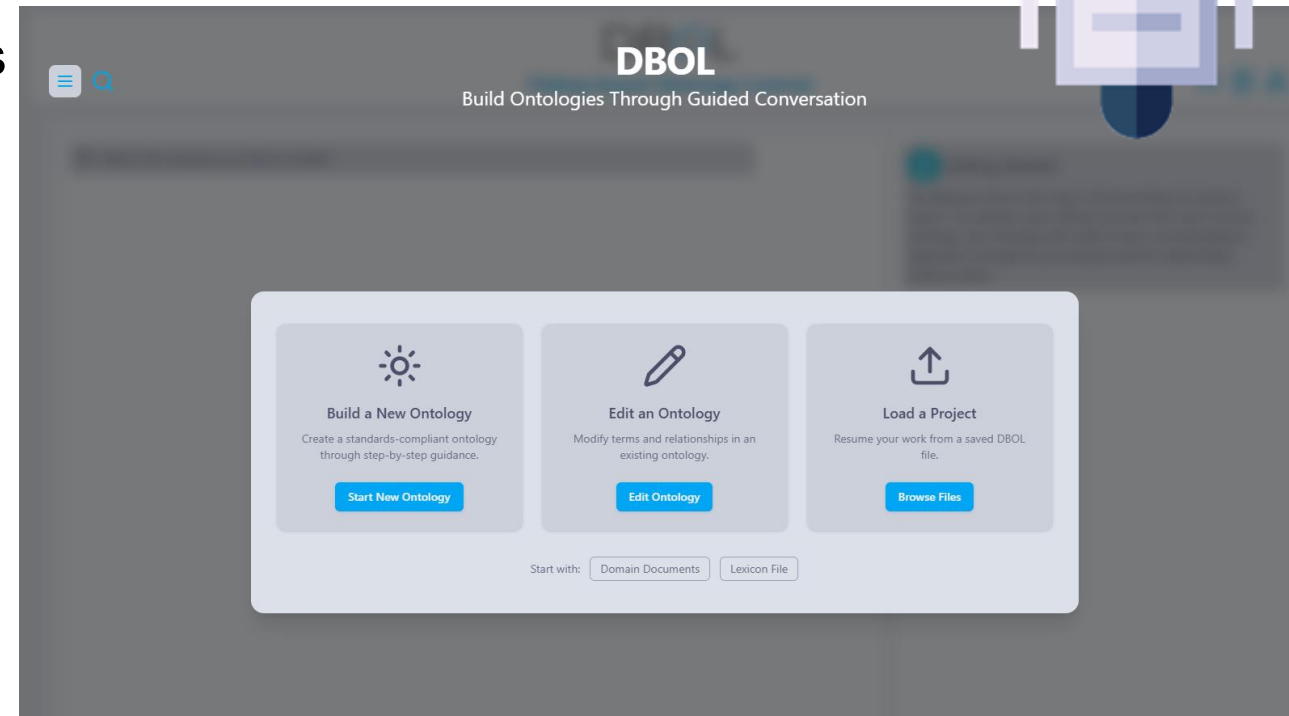


Edit elements of existing ontologies



DBOL – Dialog Based Ontology Learning

- Uses the latest advances in AI for quality control of created ontologies
- Dialog with SME to resolve ambiguity of terms / Verify the accuracy of ontology
- SME never edits the ontology directly
 - **Valid in – valid out**
 - **Enforced inclusion of relations according to design patterns**
- Produces compliant ontologies that can be imported into ontology repository
 - Maximize the reuse of existing terminology
- DBOL incorporates ontology development strategies/best practices/design patterns





System Features

DBOL
Dialog-based Ontology Learner

something else

Which best describes how a Radar System Specification exists throughout time?

Select One:

Persists or exists through time

Happens, unfolds, or occurs over time

An unchanging state

The exact beginning or end of some process, or a boundary between two processes

Which of the following options best describes a Radar System Specification?

Select One:

A property of something

Something that can have properties

Information of some sort

Content or a pattern that can be shared

This information, content, or pattern can most accurately be described as:

Select One:

Describing how to accomplish a goal

Prescribing something

Describing something

Identifying something

Another type of information

Type a message and press Enter to talk to the DBOL

Domain: Radarsystems

Term: Radar System Specification
(informal description) description of the system's design and its functions

Need help deciding?

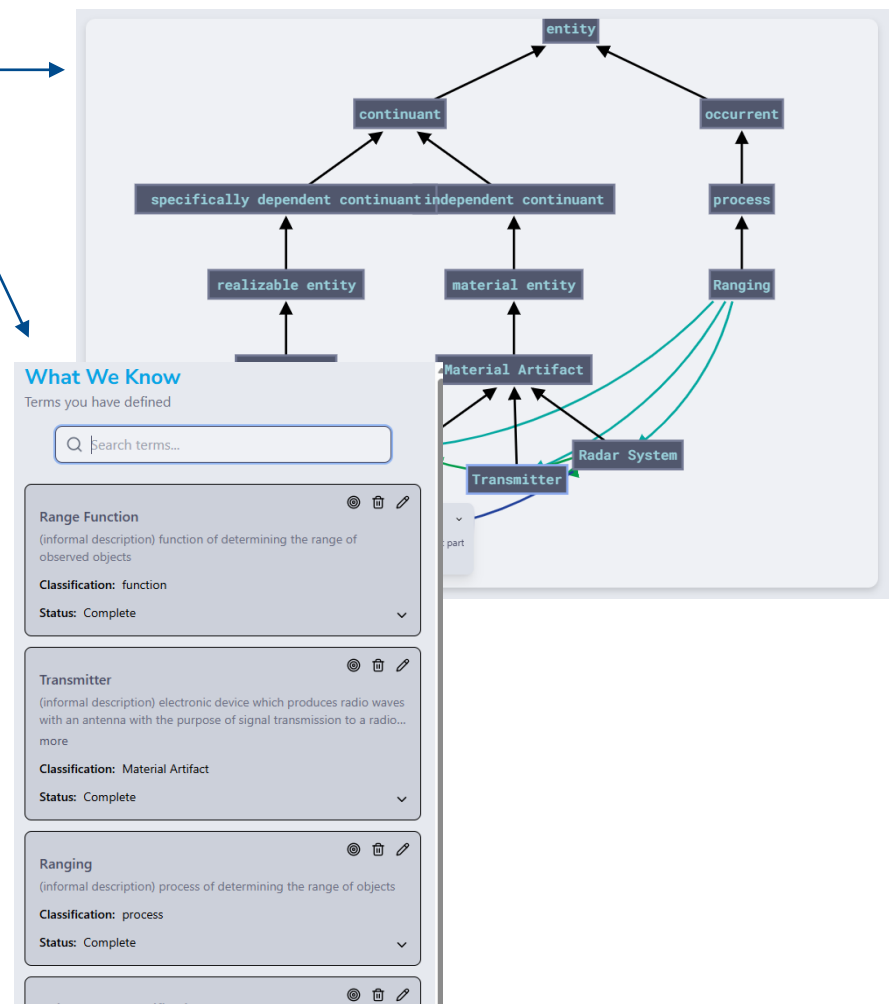
Describing how to accomplish a goal:
These **describe how to accomplish a goal**: a plan or algorithm of some sort, or a regulation that authorizes some action, or a military plan.

Prescribing something:
These **prescribe something**: what language to use, what size something should be, what kind of performance of a person is expected.

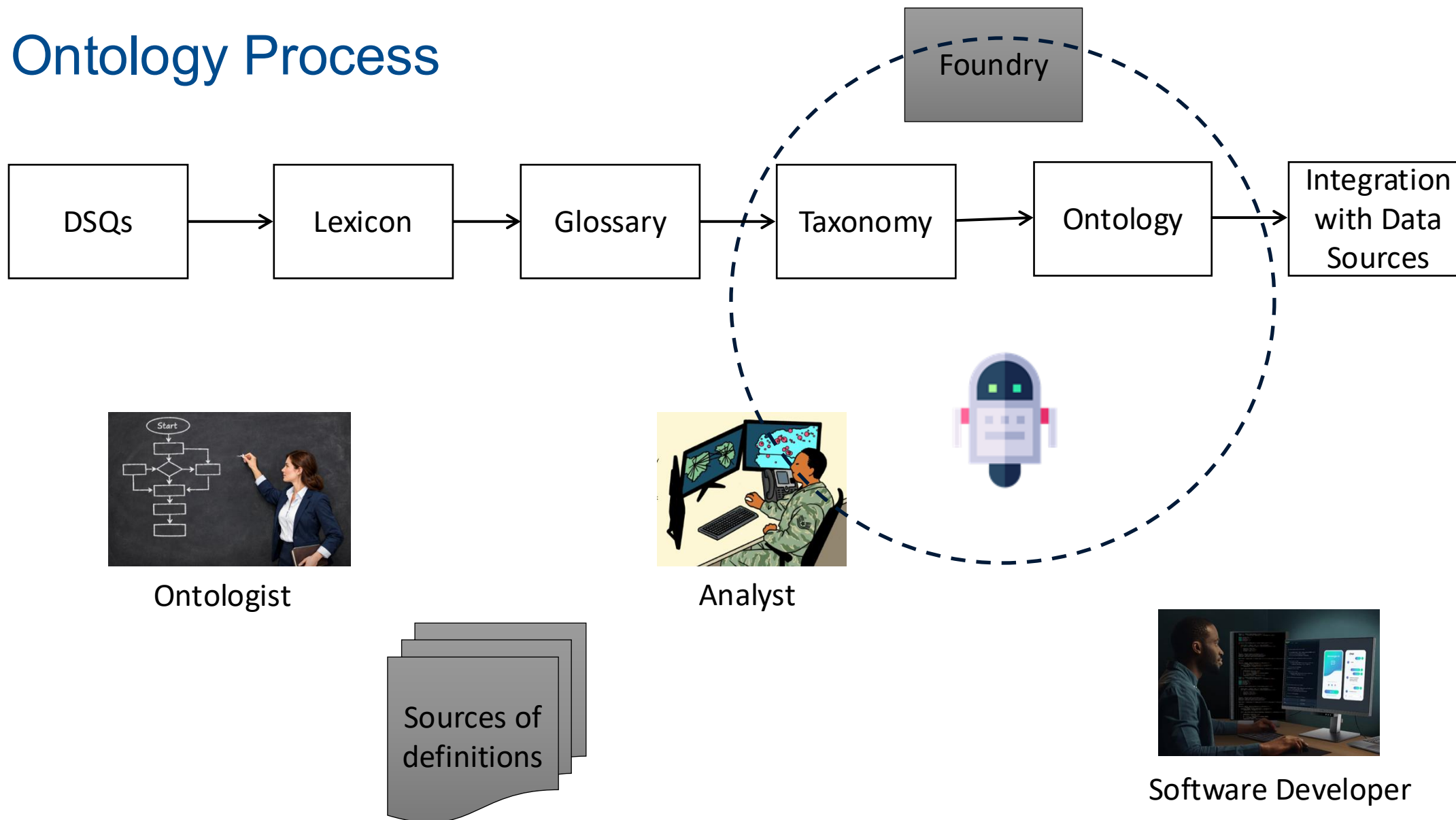
Describing something:
These **describe something**: the information in a weather report or a news story.

Identifying something:
These **identify something**: the name of a person or a vehicle identification number.

Another type of information:
There are plenty of other kinds of information like religions or ideologies.



Ontology Process





DEMO



DISCUSSION



Some of the planned features

- Support all CCO relations
- Improved dialog flow, visualization, navigation
- Implementation of workflows for specific types of use cases
- Support for design patterns
- Use dated/non-compliant ontologies as sources
- Scale up to support larger ontologies